

LUKAS RETSCHMEIER

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I am a PhD candidate in computer science at the University of Copenhagen, supervised by Prof. Rasmus Pagh. My background combines algorithm design with more than seven years of professional experience as a software developer and IT consultant. My research has resulted in publications at leading computer science conferences.

In industry, I have been responsible for technical leadership, solution design, and the implementation of large-scale software systems. I have extensive experience with modern software development practices, tools, and architectures.

Technologies: Java + Spring, AngularJS, Docker | *Additionally:* Python, C++

EDUCATION

University of Copenhagen

Doctor of Philosophy (PhD), Computer Science
Algorithms under Differential Privacy, Data Science

23 – expected 28/09/26

Supervisor: *Prof. Rasmus Pagh*

Research stay abroad at Boston University

Technical University of Munich

Master of Science, Computer Science (GPA 1.6)

19 – 23

Supervisor: *Paloma Thomé de Lima*

Exchange semesters at Universitetet i Bergen and IT University Copenhagen

Cooperative State University Mannheim

Bachelor of Science, Computer Science (GPA 1.4)

15 – 18

Supervisor: *Prof. Karl Stroetmann*

in cooperation with Atos Information Technology and Siemens Professional Education

PROFESSIONAL EXPERIENCE

Atos Information Technology, PLM Consultant, *Selected Projects:*

15 – 23

Home Appliances Industry — Technical Lead [Blog Post](#)

21 – 23

Led the development of a system for real-time monitoring and error detection of thousands of production orders by streaming and consolidating production-line data across heterogeneous databases and factories. 100+ internal users per month, including key decision-makers.

Technologies: Full-stack development, *Java* (with *Spring*) as backend, *AngularJS* front-end – microservice-based, SQL

Automotive Industry — PLM Consultant

19 – 21

Developed key components of an after-sales platform, with a focus on implementing a toolchain for validating, converting, and rendering media files.

Technologies: *Java*, *Docker*, *Siemens Teamcenter*, *IBM MQ*

Automotive Industry — Junior PLM Consultant

18 – 19

Containerized Teamcenter on Linux and used it to evaluate the error-prone process of directly upgrading Teamcenter from version 10 to 12. Significant time-savings because containerization allowed us to quickly try again with other parameters.

Technologies: *Java*, *Docker*, *C++*, *Siemens Teamcenter*

Additional responsibilities: mentoring of students, support of bid processes, experimental PoCs (e.g. Teamcenter containerization, in the cloud)

Academic Employment

Teaching: Discrete Structures (21), Algorithms and Data Structures (23, 24, 25)

External Examiner for computer science, Denmark 26 – present

LANGUAGES & CERTIFICATIONS

German (native); **English** (C1); **Danish** (C1, Studieprøven); **Norwegian** (intermediate, B1)

ITIL®3 Foundation (APMG), Business English Level 2 (LCCI)

PUBLICATIONS

- [1] Jacob Imola, Rasmus Pagh, and **Lukas Retschmeier**.
“*Lower Bounds for Private Graph Problems via Reconstruction Attacks*”. In: PODS 2027.
- [2] Lukas Geis, Alexander Leonhardt, Johannes Meintrup, Ulrich Meyer, Manuel Penschuck, and **LR**.
“*Revisiting a Successful Reduction Rule for Dominating Set*”. In: ALNEX 2026.
- [3] Rasmus Pagh, **Lukas Retschmeier**, Hao Wu, and Hanwen Zhang.
“*Optimal Bounds for Private Minimum Spanning Trees via Input Perturbation*”. In: PODS 2025.
—**Note:** Improves *Faster Private Minimum Spanning Trees* (TPDP 2025) by subset of authors.
- [4] Joel Daniel Andersson, **Lukas Retschmeier**, Rasmus Pagh, and Boel Nelson.
“*Private Lossless Multiple Release*”. In: ICML 2025.
- [5] Christian Janos Lebeda and **Lukas Retschmeier**.
“*The Correlated Gaussian Sparse Histogram Mechanism*”. In: FORC 2025.

Under Submission

- [6] Anders Aamand, Rasmus Pagh, and **Lukas Retschmeier**.
“*Optimal Bounds for Differentially Private All Pairs Shortest Paths*”.
- [7] Paloma Lima and **Lukas Retschmeier**.
“*A Linear Kernel for Planar Semitotal Dominating Set*”.

COMMUNITY SERVICE

Reviewer — ICML (25), TMLR (25, 26), RANDOM (25) — SEA (26)

PC Member — TPDP (26)

HONORS

Best graduate in informatics (*DHBW Mannheim*)

A first prize in the piano competition *Jugend Musiziert*

OTHER INTERESTS

Piano and classical music, (winter) bathing, volleyball